

Campus Facility Reservation System

Project Attribute	Value	Project Attribute	Value
Project Name	Campus Facility Reservation System	Document Version	1.1.0
Project Sponsor	Mishal Rai	Target Audience	Engineering Interns
Duration	6-Day Sprint	Status	APPROVED FOR SPRINT

1. Executive Summary & Objective

- **The Context:** Campus facilities are often reserved but left unused because students fail to show up.
- **The Business Objective:** Create a fair booking system using refundable token deposits and cancellation penalties.
- **The Engineering Objective:** Learn how to prevent double-booking and apply time-based business rules.

2. Recommended Technology Stack

- **Frontend:** JavaScript (Vanilla, React, or Vue), HTML5, CSS3
- **Backend:** Python 3.x using FastAPI or Flask frameworks
- **Database:** MySQL 8.0
- **Testing & Tools:** Postman API Desktop, Python Native Logging Framework

3. Functional Requirements

- **Schedule Calendar:** Facilities are mapped into hourly reservation slots.
- **The Booking Deposit:** Reserving a slot deducts a token deposit and marks the slot as Reserved.
- **Cancellation Rules Engine:** Refund percentages depend on how early a user cancels.
- **More than 24 Hours:** 100% refund.
- **Between 12 and 24 Hours:** 50% refund and 50% penalty.
- **Less than 12 Hours:** No refund and full penalty applied.

4. Postman API Testing & System Activity Logs

- **Postman Test Suite:** Validate that simultaneous booking attempts cannot reserve the same slot twice.
- **INFO:** Successful slot booking with student_id, slot_id, and reservation time.

- **WARNING:** Penalty applied because cancellation happened too close to start time.
- **ERROR:** Student tries to cancel a reservation that already occurred.

5. 6-Day Sprint Roadmap

Day	Phase & Deliverable	Resources
Day 1	Database Architecture for slots and bookings tables.	MySQL 8.0, DB Schema Docs
Day 2	Python Business Logic for time calculations and locking.	Python 3, FastAPI/Flask
Day 3	API Engineering for booking and cancellation endpoints.	VS Code, Postman
Day 4	Frontend Interface with visual calendar and dashboards.	React / Vanilla JS, HTML5/CSS3
Day 5	Postman double-booking verification and log testing.	Postman, Python Logging
Day 6	QA testing, bug fixes, and final code merge.	Git, GitHub Workflows

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